SPL1 Project Proposal Form, 2026

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Project Description:

Title: Advanced Multi-User Terminal Chat System with End-to-End Encryption (E2EE)

This project involves developing a secure, real-time, low-level multi-user terminal chat application operating on a **Zero-Knowledge Client-Server Relay Architecture**. The system manages concurrent user connections, payload-level message encryption, and interactive terminal session management directly using Operating System APIs without relying on high-level web frameworks or abstraction wrappers.

Key Technical Features:

- **End-to-End Encryption (E2EE):** Implements dynamic symmetric key exchange based on the Diffie-Hellman Key Exchange algorithm alongside a handcrafted stream cipher. Payload encryption and decryption occur strictly on user client endpoints.
- **Multithreaded Networking & Concurrency:** Utilizes native OS multithreading APIs (`CreateThread` / `sys_clone`) and TCP sockets (`Winsock2` / `sys_socket`) to handle non-blocking concurrent client connections and parallel execution.
- **Low-Level Custom Utilities:** Bypasses default standard C library abstractions by featuring custom-written memory management (`my_memset`, `my_strcpy`), string processing (`my_strlen`, `my_strcmp`), and console I/O routines.
- **Interactive CLI Command Engine:** Supports `/users` (active user roster), `/msg <user> <message>` (private peer messaging with sound alert), `/help`, `/clear`, and `/exit`.
- **Rich Terminal UI:** Integrates Windows Virtual Terminal Processing for ANSI color formatting and automatic local timestamps (`[HH:MM:SS]`).

Languages or Tools to be used:

- **Programming Language:** C (Systems Programming)
- **Networking Protocol/API:** TCP/IP, Winsock2 API (`ws2_32.lib`) / OS Socket API
- **Concurrency & OS API:** Win32 API (`windows.h`, `Kernel32`, `CreateThread`, Virtual Terminal Processing)
- **Compiler & Development Tools:** GCC (MinGW-w64), Visual Studio Code

Supervisor's Name: Prof. Mohd. Zulfiquar Hafiz

Signature of the supervisor: _____ **Date:**

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Before the Midterm Presentation:

I confirm that the progress is satisfactory and I am forwarding it for midterm presentation.

Signature of the supervisor: _____

Date: _____

Midterm Presentation Feedback:

Before the Final Presentation:

I confirm that the project is complete and I am forwarding it for final presentation.

Signature of the supervisor: _____

Date: _____